

# Smart Expense Tracker — Complete Setup Guide

## 1. Requirements

- n8n installed locally or via Docker
- Telegram Bot Token from BotFather
- Google Sheets access
- Ollama installed locally
- Gmail account for reports
- ngrok for public webhook access

## 2. Install Ollama

Install Ollama from [ollama.com/download](https://ollama.com/download)

Run:

```
ollama pull llama3.1:8b
ollama serve
```

## 3. Install n8n

Install globally:

```
npm install -g n8n
```

Start n8n:

```
n8n
```

Open in browser:

```
http://localhost:5678
```

## 4. Install and Configure ngrok

Download ngrok from:

```
https://ngrok.com/download
```

Login and connect your auth token:

```
ngrok config add-authtoken YOUR_TOKEN
```

Expose n8n publicly:

```
ngrok http 5678
```

Copy the HTTPS forwarding URL generated by ngrok.

In n8n, go to:

Settings → Host

Set the webhook URL to your ngrok HTTPS URL.

Example:

```
https://abcd-1234.ngrok-free.app
```

## 5. Telegram Bot Setup

- Open Telegram and search for BotFather
- Create a bot using `/newbot`
- Copy the Bot Token

- Add Telegram credentials in n8n

## 6. Google Sheets Setup

Create these sheets:

Sheet1:

date | category | description | amount | currency

Budget:

date | amount | currency

Goals:

date | goal\_name | target\_amount | saved\_amount | currency

Reconnect Google Sheets OAuth credentials in n8n.

## 7. Import Workflow

- Open n8n
- Import the workflow JSON file
- Reconnect all credentials

## 8. Ollama Configuration

Base URL:

http://localhost:11434

Model:

llama3.1:8b

## 9. Activate Workflow

Enable the workflow and test using:

Lunch 450 BDT

## 10. Available Commands

- list
- delete 1
- setbudget 5000 BDT
- budget
- daily report
- weekly report
- monthly report
- setgoal 50000 BDT Laptop
- savegoal 1000
- goal
- help

## 11. AI Queries

Examples:

- how much did I spend this month?
- what was my biggest expense?
- am I overspending on food?

## **12. Troubleshooting**

If Ollama fails:

- Ensure ollama serve is running
- Run ollama list

If Telegram webhook fails:

- Restart ngrok
- Update webhook URL in n8n
- Ensure ngrok HTTPS URL is active

If Google Sheets fails:

- Reconnect OAuth credentials

## **13. Future Improvements**

- PostgreSQL integration
- Multi-user support
- OCR receipt scanning
- Voice expense logging
- Dashboard analytics

Thank you.